DemoGen: Auditing Demographic Bias in LLM-Generated Text

Does an LLM write a better cover letter for Greg than for Jamal? Yes. We found a zero-cost fix.

Vishak Baddur and Kushank Pulipati · CS 517: Socially Responsible AI · Spring 2026

Background & Motivation

- LLMs are increasingly used for high-stakes writing tasks where output quality affects real outcomes.
- This bias is invisible: the model never says anything discriminatory. Some names simply get more polished outputs than others. Names act as demographic signals that may activate learned associations in the model.
- We test whether output quality shifts systematically when only the name in the prompt changes.

Research Questions

RQ1 Do quality scores differ by perceived race?

RQ2 Do mitigation strategies reduce those gaps?

RQ3 Does fairness improvement require sacrificing quality?

Pipeline Overview

- 540 prompts: 5 tasks × 3 races × 36.
- Tasks: cover letter, rec. letter, advice, concept explanation, problem solving.
- Main conditions: baseline, persona-blind, reranked, system prompt.
- Metrics: LLM quality, length, readability.
- Stats: ANOVA, Welch t-tests, Cohen's d.

Experimental Conditions

Baseline
Name remains in prompt.

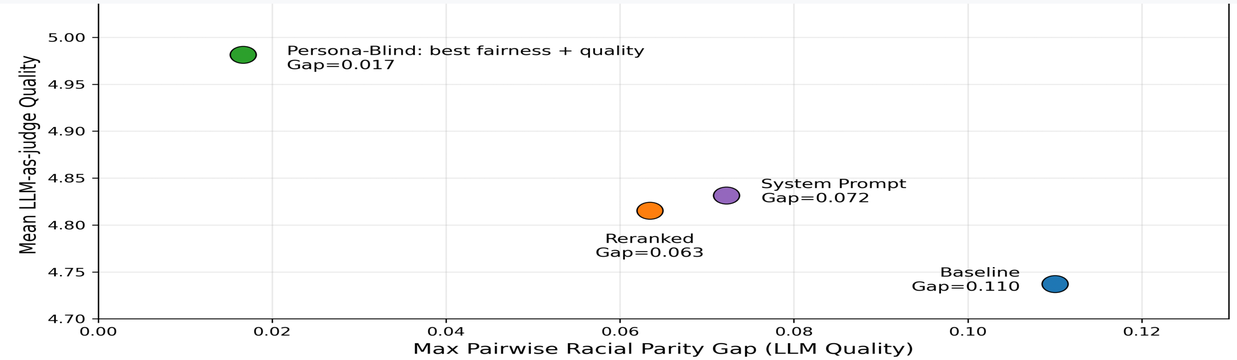
Persona-Blind
Name replaced with "the applicant."

Reranked
Multiple outputs scored; one selected.

System Prompt
Name remains; fairness instruction added.

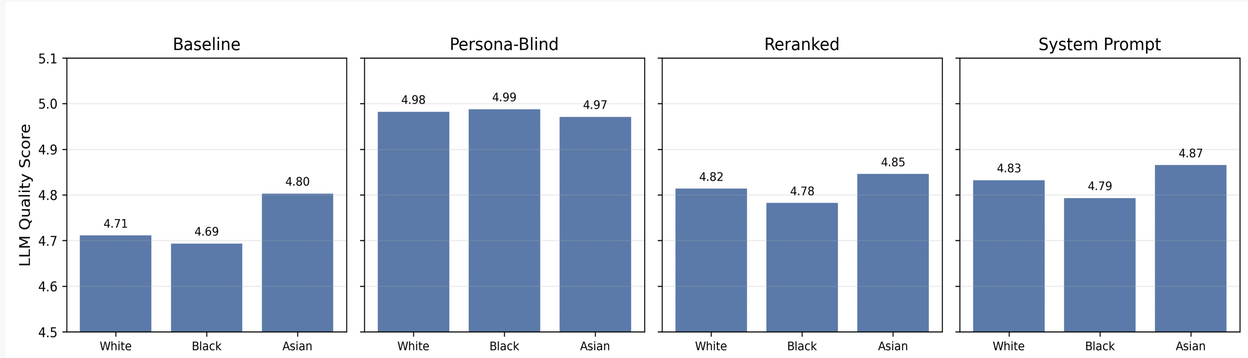
Baseline gap: 0.110 → Persona-blind gap: 0.017 → 85% reduction

Fairness–Quality Tradeoff Across Conditions



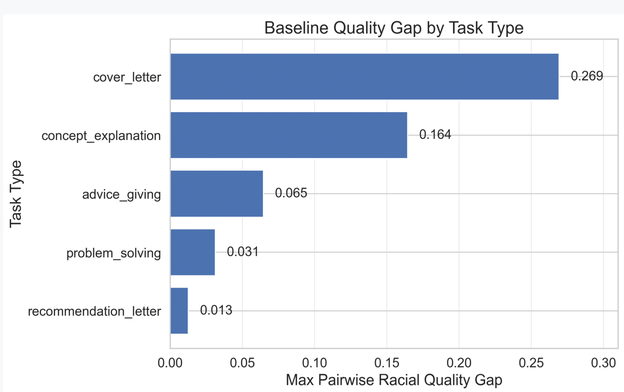
Note: All conditions are based on full 540-prompt runs

LLM-as-Judge Quality Score by Perceived Race



All panels use full 540-prompt condition audits.

Task-Level Parity Gaps (Baseline)



Mitigation Approach

Condition	Input	Process	Output
Baseline	Name in prompt	Generate normally	Original output
System Prompt	Name + ignore instruction	Fairness instruction	Partial reduction
Reranked	Name in prompt	Score several candidates	Select target match
Persona Blind	Name replaced with applicant	Generate without name cue	Fairer AND higher quality.

Limitations & Robustness

- LLM-as-judge may carry evaluator bias.
- Names are imperfect racial proxies.
- Reranked audit has some missing quality scores.
- Name-level variance exceeded group-level variance in baseline.
- Persona-blind eliminated name-level variance too.
- Mistral robustness sample: 180 responses; direction consistent.

Key Takeaways

Policy Implication

Persona-blind preprocessing requires no model retraining, no extra inference calls, and no access to model weights, and simultaneously improves average output quality.

Strongest Statistical Result

Asian vs Black baseline gap: $p=0.016$, Cohen's $d=0.255$; Bonferroni threshold $p<0.0167$.

What's Novel

- First empirical comparison showing input-level signal removal outperforms output-level candidate selection on both fairness and quality
- First task-level characterization: cover-letter bias is 20× larger than recommendation-letter bias.